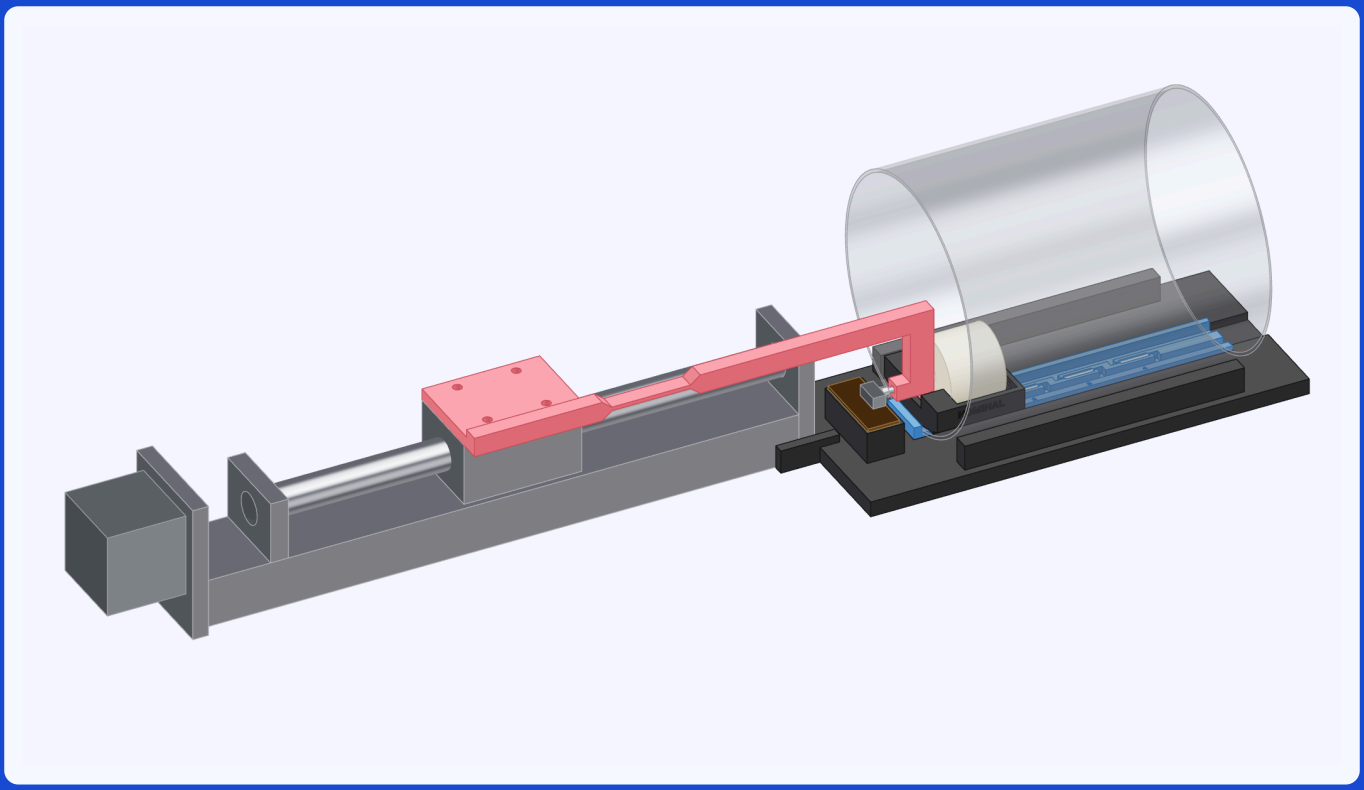
HOME SODA MACHINE

Reed sweep rig

Build and commission the bench fixture that measures where the Aero float's magnet closes and opens the carbonator's level switches.



BUILD GUIDE

Contents

Follow the sections in order to print, assemble and commission the rig before taking calibration readings.

SECTION	PAGES
Before assembly Rig overview, bench kit and print plates	3-5
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Set the float and home Sled, roller, clearances and striker	11-14
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Check the wiring Meter checks, USB inputs and motor current	18-20
Run the rig Verify direction and home, record sweeps and set reed heights	21-23

READING THE DIAGRAMS

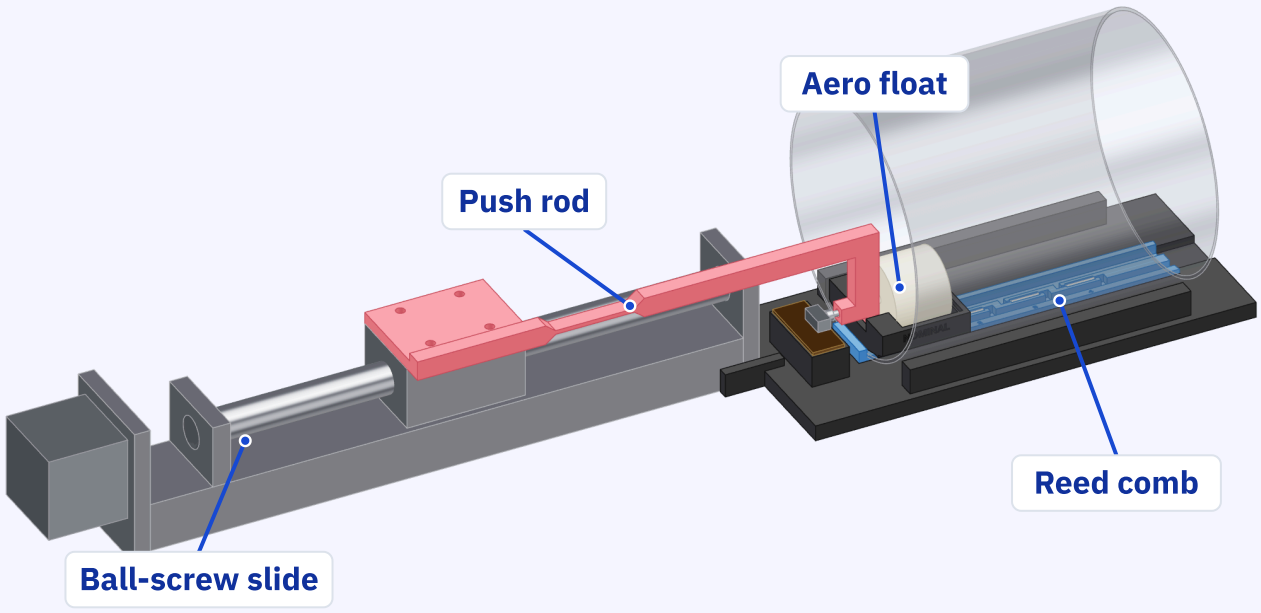
In the assembly views, coral highlights the part being fitted or the direction of motion. Brass marks heat-set inserts. Blue marks a datum or gauge. A translucent tube reveals the parts inside.

Use page numbers in the website reader. Open the downloaded PDF for bookmarks and reference links. Print single-sided on Letter at actual size.

BEFORE ASSEMBLY

How the rig works

The slide moves the Aero float through the dry tube. The controller logs where each reed switch closes and opens as the magnet passes.



The tube rim at the slide end seats against the comb tab. That rim represents the bottom of the upright carbonator. Positions increase into the tube.

The sled sets the gap

NEAR, **NOMINAL** and **FAR** place the same float progressively farther from the reeds. The sled's runners ride on the bore.

The switch sets home

The striker trips the switch as the sled moves toward the mouth. The trip sets the controller's zero. The clearance from the tab puts that zero in the tube's coordinate system.

SET UP ON WOOD OR PLASTIC

Steel can distort the float's magnetic field. Use the specified stainless fasteners near the tube mouth and keep loose steel tools and fasteners a hand's width from the tube while taking readings.

BEFORE ASSEMBLY

What to have on the bench

Lay out the motion parts, controller and test pieces before heating the insert tool.

USE	PART
Slide	Befenybay BE069-2 ball-screw stage with its NEMA 17 motor and lead
Drive and power	BIGTREETECH TMC2209 V1.3, ZXXP stepper driver carrier and BTF-LIGHTING 12 V adapter with its female barrel connector
Logic	DORHEA ESP32 with screw-terminal breakout and a USB-C data cable
Home	R REIFENG endstop board and lead
Test pieces	Bare carbonator tube, finished Aero float, and the reeds you intend to install
Fasteners	Three ruthex M3 heat-set inserts, two 316 stainless M3 × 8 switch screws, one 304 stainless M3 × 12 striker, four M4 carriage screws and washers
Wire	Thin insulated signal and return wires for the comb, plus Dupont jumpers for the carrier

Bench tools

Heat-set tip, soldering iron, fine pliers, cutters, hex keys, multimeter, calipers and smooth gauge pins or drill shanks. Set out a bank card for the striker and an insulated tool for the driver's trimmer.

Part links in the tool ledger

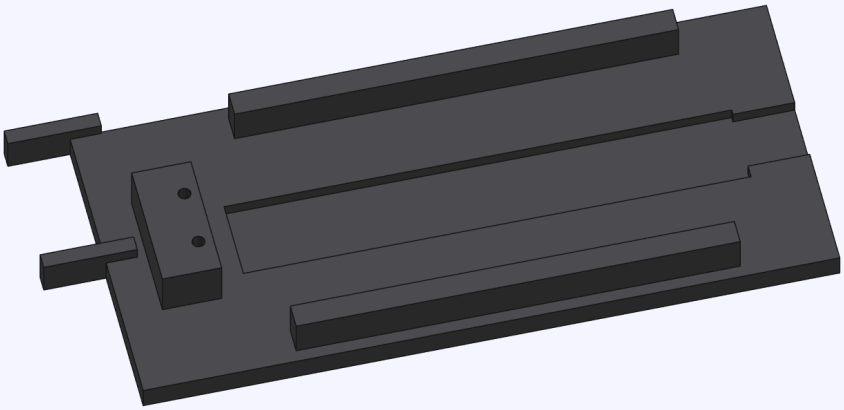
BEFORE ASSEMBLY

Dry-fit the slide and switch on the printed cradle. Check that the slide end plate seats squarely and the switch mounting holes line up with the insert holes.

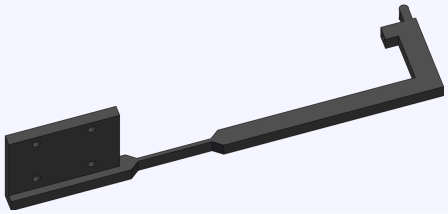
BEFORE ASSEMBLY

Print the cradle and small parts

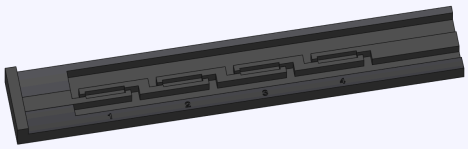
Print both prepared plates for your printer in black PET-GF. The projects include the part orientations and support settings.



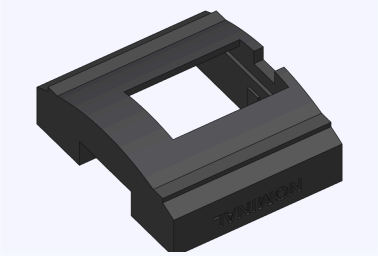
Cradle plate. The cradle lies flat, with the comb pocket, switch pad and keepers up.



Push rod. The pad stands on edge.



Reed comb. Pockets and wire grooves face up.



Sleds. Print NEAR, NOMINAL and FAR with the runners facing up.

Small-parts plate

The rod, comb and all three sleds share one plate. Both prepared plates print without supports.

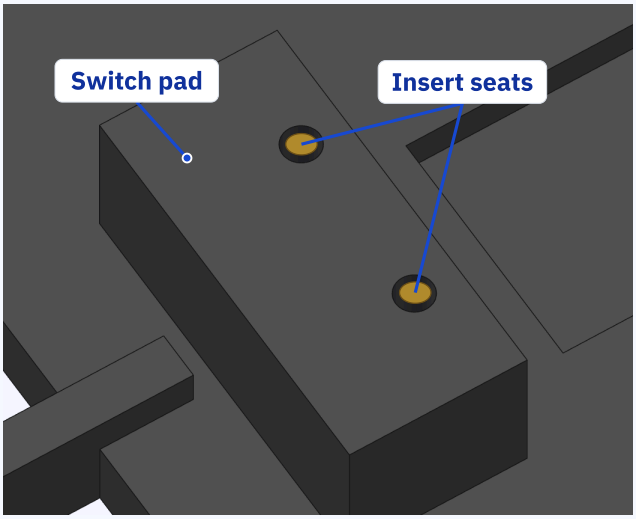
Mark1 and Mark2 print files

BEFORE HEAT-SETTING

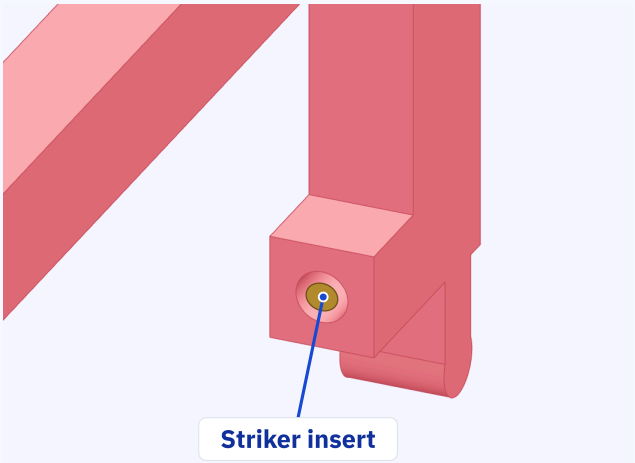
The comb drops fully into its pocket. Clear stray strands from the reed pockets, sled slots and gauge windows. Keep the sleds' runner edges square.

Heat-set the inserts

Fit the switch inserts in the cradle pad and the striker insert in the end of the push rod's finger.



The holes on the raised pad take the switch screws.



Stand the rod so you can press squarely into the finger.

1. Check that the home switch board's mounting holes line up with the holes in the pad. Clear stray filament from the insert holes.
2. Press each insert squarely into its hole with the heat-set tip. Stop when the insert is flush with the surrounding face.
3. Let the plastic cool. Hand-start the matching screw in each insert.

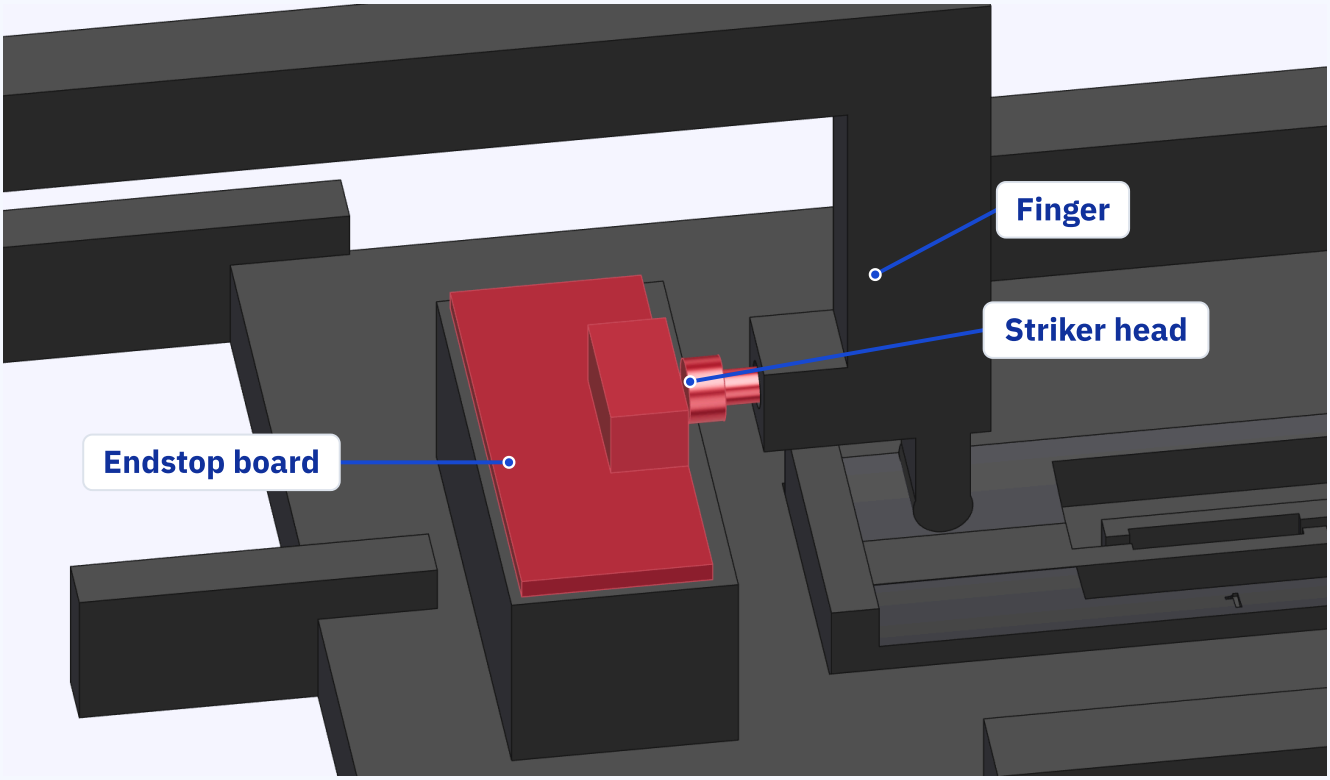
CHECK THE SEATING FACES

The switch board can lie flat and each screw threads by hand. Clear any raised plastic around the insert before fitting the board.

ASSEMBLE THE RIG

Fit the switch and striker

Point the switch lever toward the tube and the striker head toward the lever.



The home switch board mounts through the holes beside the switch body. The striker head presses the lever as the sled approaches the tube mouth.

1. Set the board on the cradle pad with the lever toward the tube. Fasten it with the 316 stainless M3 × 8 screws.
2. Thread the 304 stainless M3 × 12 striker screw most of the way into the finger by hand. Adjust it after coupling the sled.

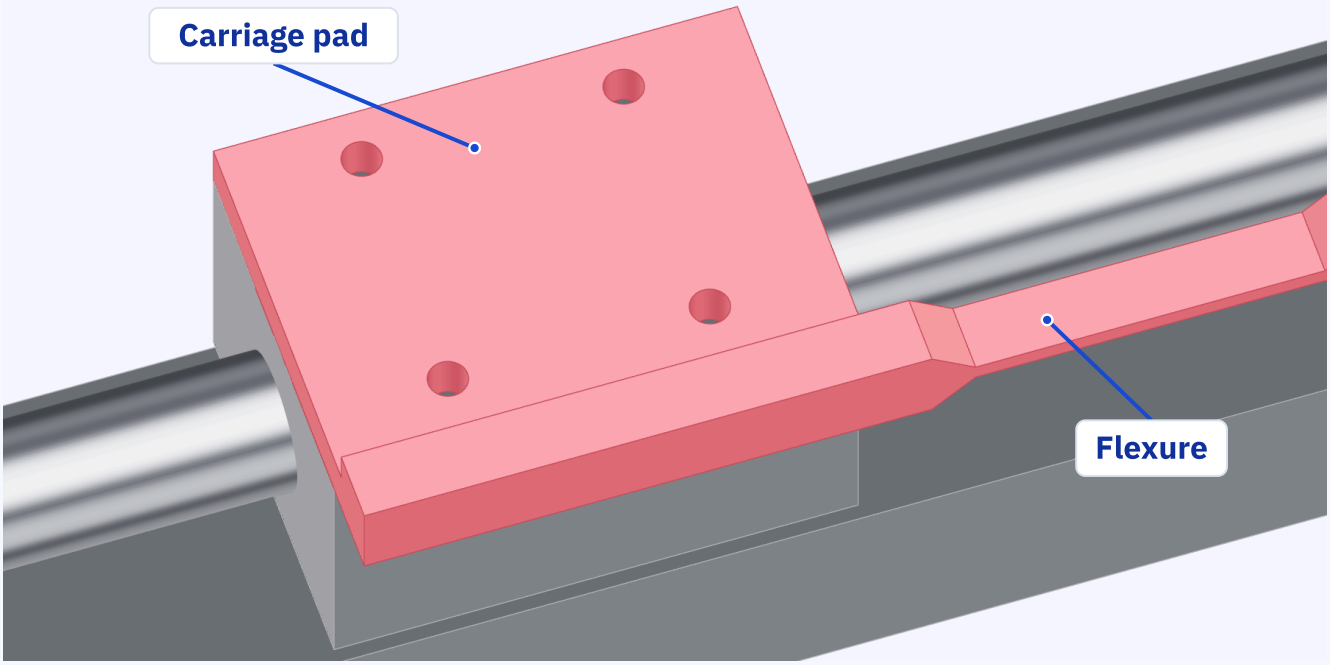
CHECK THE SWITCH BY HAND

The board sits flat and stays put. The lever moves freely and returns when you release it.

ASSEMBLE THE RIG

Bolt the rod to the carriage

Set the push rod's pad on top of the carriage. The bar runs beside the carriage and the roller leg points down at the tube end.



The thin section beside the pad flexes. Lift the rod at the tube end just enough for the roller to clear the sled.

1. With the stage unpowered, turn the coupling by hand until the carriage is near the middle.
2. Try an M4 screw by hand in a bare carriage hole. Compare its available thread depth with the screw's projection through the pad and washer. Add washers or use a shorter screw if it would bottom.
3. Set the pad flat on the carriage and fasten it through all four holes with the M4 screws and washers.

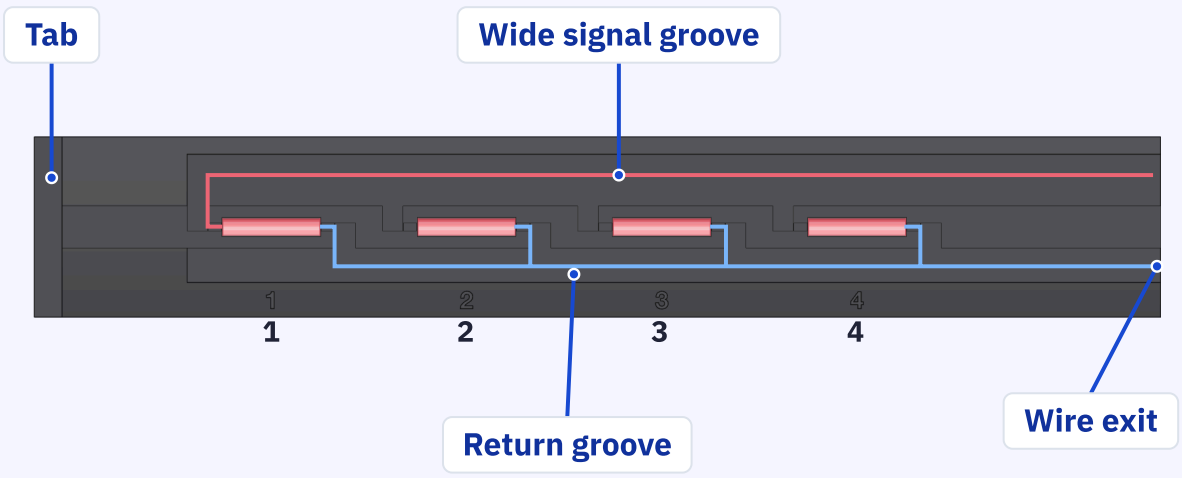
CHECK THE CARRIAGE

The pad sits flat and the screws clamp it before they bottom out. The roller returns when you release the rod.

ASSEMBLE THE RIG

Wire the reed comb

Use the reeds you plan to install in the machine. Give each reed its own signal wire and join the opposite leads to one common return.



Count stations from the tab. Low and high are the middle pair. Coral shows one signal wire. Blue shows the common return.

1. Seat each glass body against the end of its pocket nearest the tab.
2. Grip the lead with pliers between the glass and the bend. Bend the lead nearest the tab through its slit into the wide groove. Solder on a thin signal wire labeled with the station number.
3. Support the other lead the same way. Bend it into the narrow groove and solder it to the common return. Trim the leads inside their grooves and route every wire out the end opposite the tab.

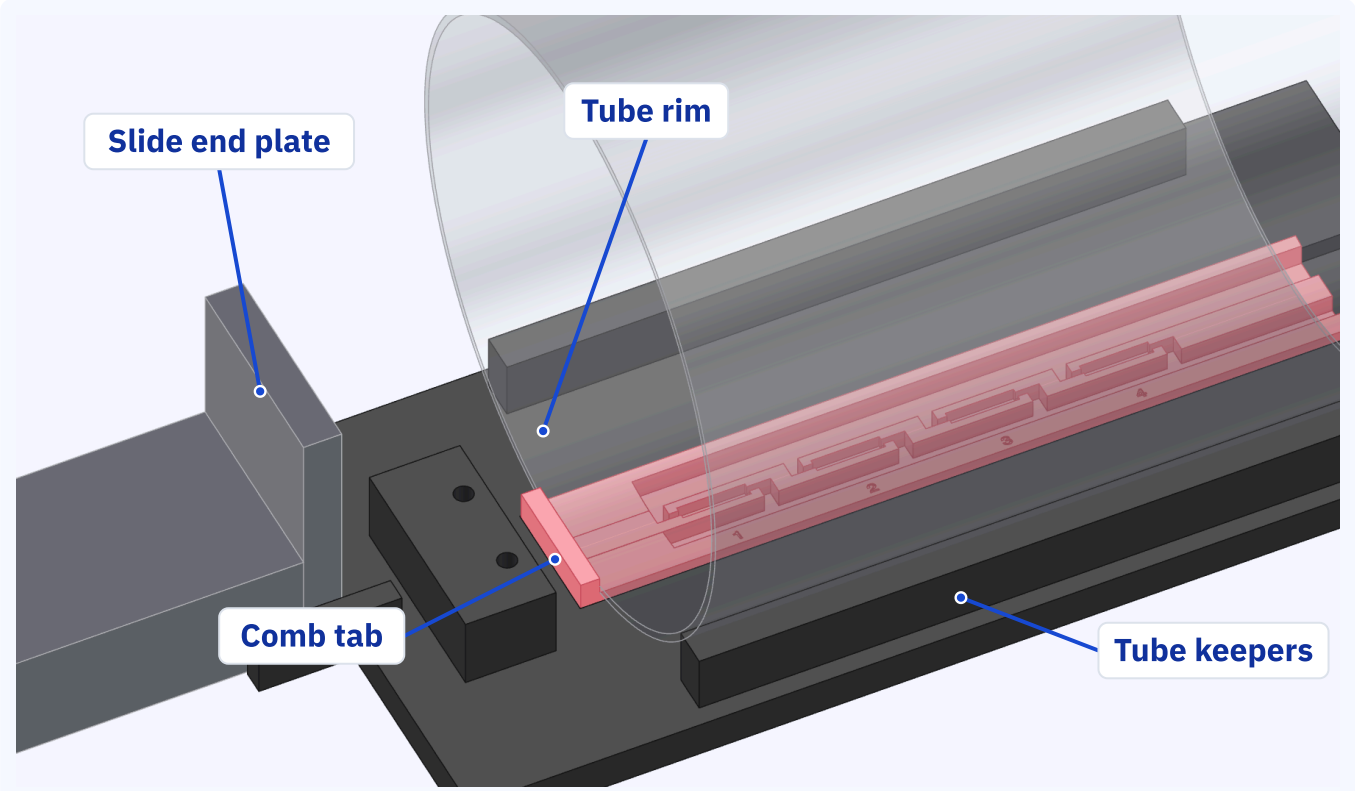
CHECK CLEARANCE AND CONTINUITY

Keep glass, wire and solder below the flat surface that supports the tube. Check continuity between each signal and the return as you pass the float's magnet over the reed. The contact should close and reopen.

ASSEMBLE THE RIG

Seat the comb, tube and slide

Set the cradle and slide on the same flat wood or plastic work surface.



The cradle cheeks align the slide with the tube. The comb tab sets the tube's axial position.

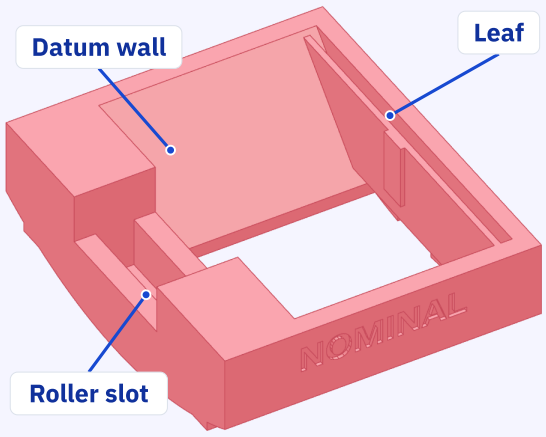
1. Drop the comb into its pocket with the tab toward the switch pad. Lay the wires in the trough at the far end.
2. Lower the bare tube between the keepers with its weld seam up. Slide the rim against the tab.
3. Fit the end of the slide opposite the motor between the cradle cheeks. Seat the end plate against the cradle and align the rod with the tube mouth.

CHECK THE SEATING

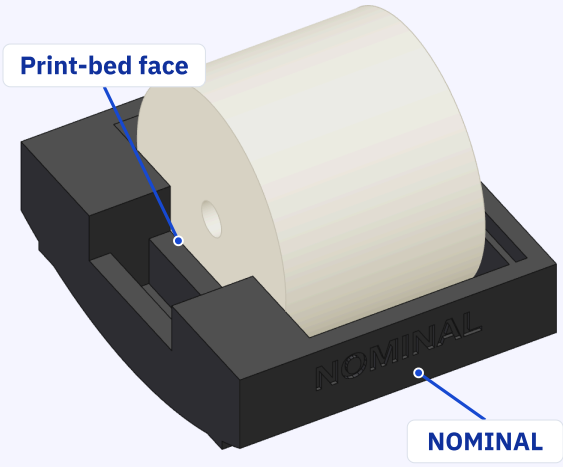
The tube rests on the comb without rocking or pinching a wire. The rim touches the tab and the end plate sits squarely between the cheeks.

Seat the float in the sled

Start with the NOMINAL sled. The float's print-bed face rests against the datum wall beside the roller slot.



Empty. The datum wall is between the slot and the vee.



Seated. The float lies on its side. The leaf holds its bottom face against the datum wall.

1. Hold the print-bed face flat against the datum wall, with the guide bore parallel to the sled's runners.
2. Press the float into the V-shaped seat until the far face passes the leaf and the leaf springs back.

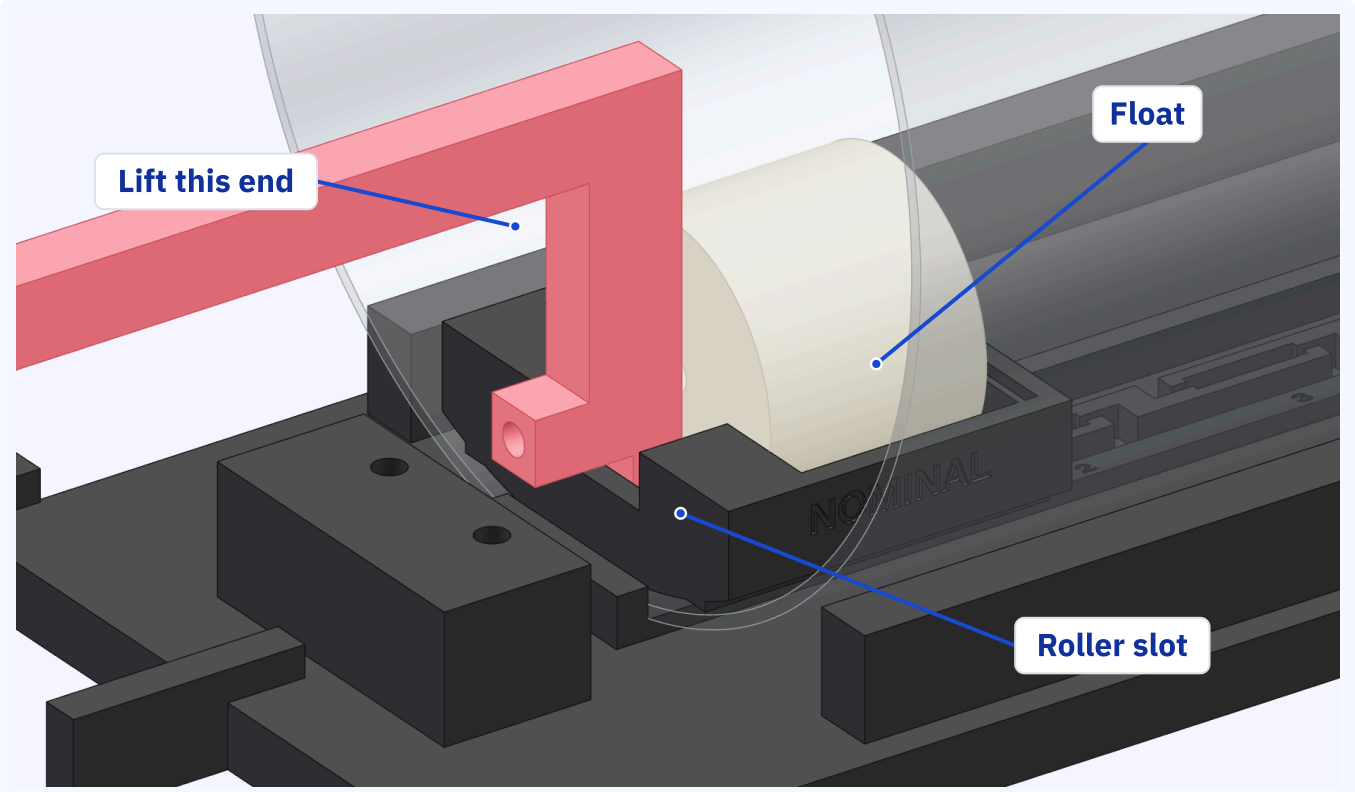
CHECK THE SEAT

The float sits in the V with its print-bed face against the datum wall. The roller slot stays clear and the float stays seated when you turn the sled over.

SET THE FLOAT AND HOME

Couple the sled to the rod

Insert the loaded sled slot first from the end opposite the slide. Couple the roller to the sled at the tube mouth.



The rod leg drops into the slot ahead of the float. The slot's front wall hides the roller once it seats.

1. With motor power disconnected, feed the sled toward the mouth until the slot is under the rod leg.
2. Lift the far end of the rod just enough to clear the sled, then lower the roller fully into the slot.
3. Turn the slide coupling by hand to move the sled away from the tab and back. Watch both runners and the roller.

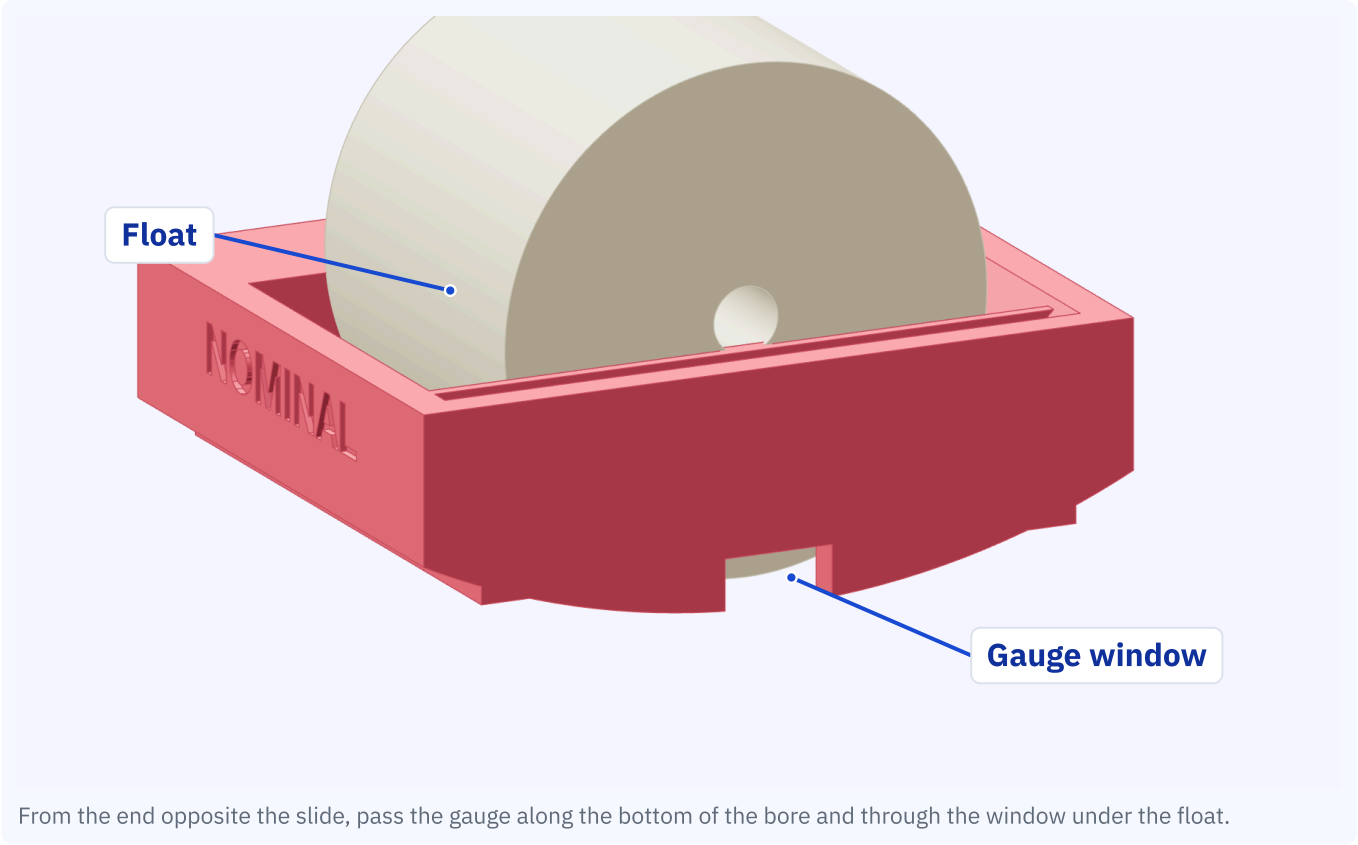
CHECK THE COUPLING

The roller stays seated when you reverse direction. Both runners ride on the bore and the float clears the wall. If the sled binds, check the rod height, cradle seating and sled orientation.

SET THE FLOAT AND HOME

Measure the float gap and roller play

Measure the clearance under the float and the play between the roller and its slot for each sled you use.



From the end opposite the slide, pass the gauge along the bottom of the bore and through the window under the float.

1. Find the largest gauge pin or smooth drill shank that passes under the float without lifting it or rocking the sled.
2. Record the measured gap with the sled name. Compare with that sled's `float_to_wall_gap_mm` in the rig datums.
3. Remove the gauge and sled. Measure the slot along the travel axis and subtract the roller diameter. Record that difference as roller play, then recouple the sled.

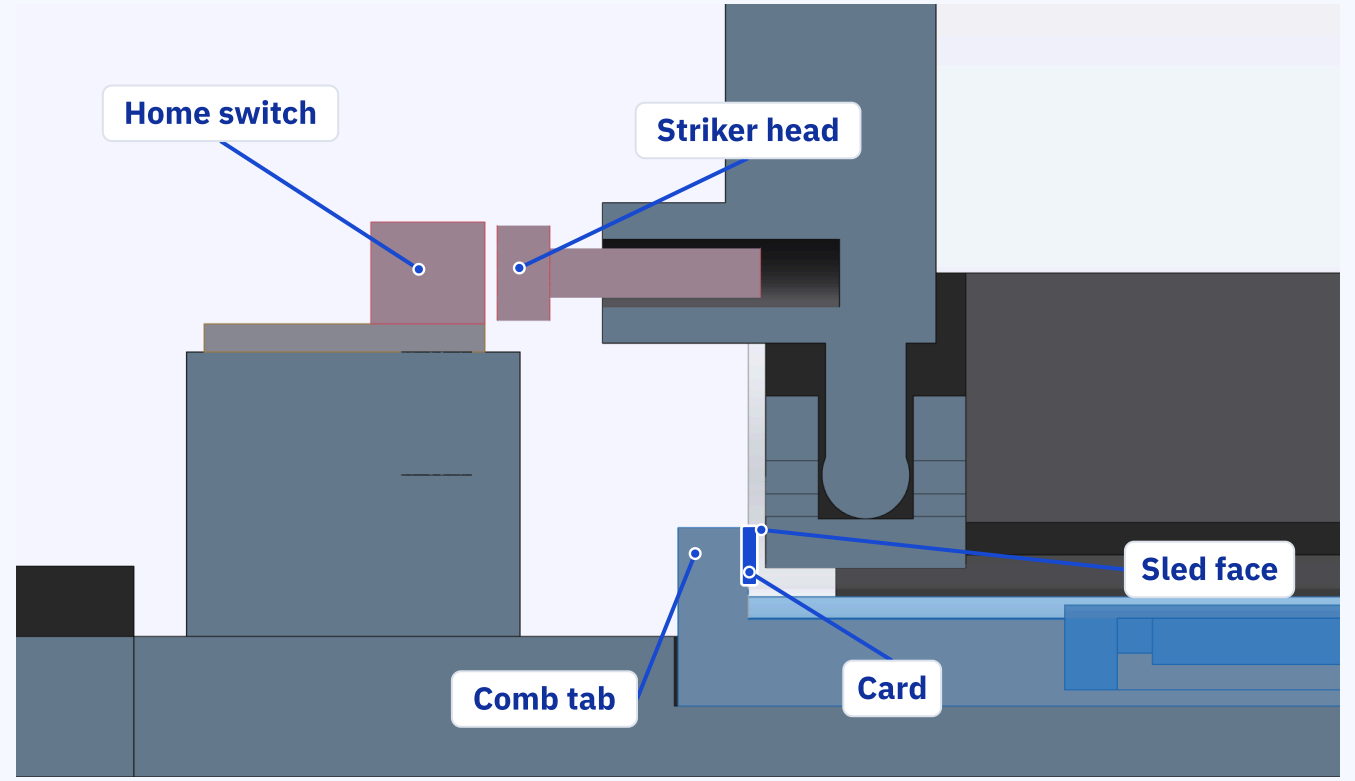
CHECK THE RUNNING GAP

The gauge enters without force and the float clears the tube through the hand sweep. The log reader uses the measured roller play to correct the inward leg.

SET THE FLOAT AND HOME

Set the striker at the tab

The tab locates the tube rim. Set the switch to trip before the sled's leading face reaches that same stop.



Side section at the tube mouth. The blue card sets the gap between the sled face and the comb tab at the switch trip.

1. Leave the motor unpowered. Measure a bank card's thickness at the spot you'll use as the spacer. Place the card between the sled's near face and the tab. Turn the coupling by hand until the sled just grips it.
2. Back the striker screw out toward the switch until the lever clicks. Draw the card out.
3. Move the sled into the tube until the switch releases, then bring it back slowly. Stop as soon as it clicks.

CHECK THE GAP AT THE CLICK

The switch clicks with a card's thickness still between the sled and the tab. Keep the measured thickness for the controller's `tab <mm>` setting.

WIRE THE CONTROLLER

Match the driver to the carrier

Match the driver's enable and motor-supply corners to the carrier before seating the TMC2209.



The driver's EN and motor supply pins sit at the same end, on opposite rows. Match both to the carrier's markings.

1. Disconnect USB and the motor adapter. Locate EN and the motor supply pin marked VS or VM on the driver.
2. Set all carrier configuration switches to OFF for standalone STEP/DIR. Confirm that both microstep inputs are low, the setting expected by the rig firmware.
3. Compare both socket rows with the pinout. Set the driver aside until the wiring and carrier voltage checks are complete.

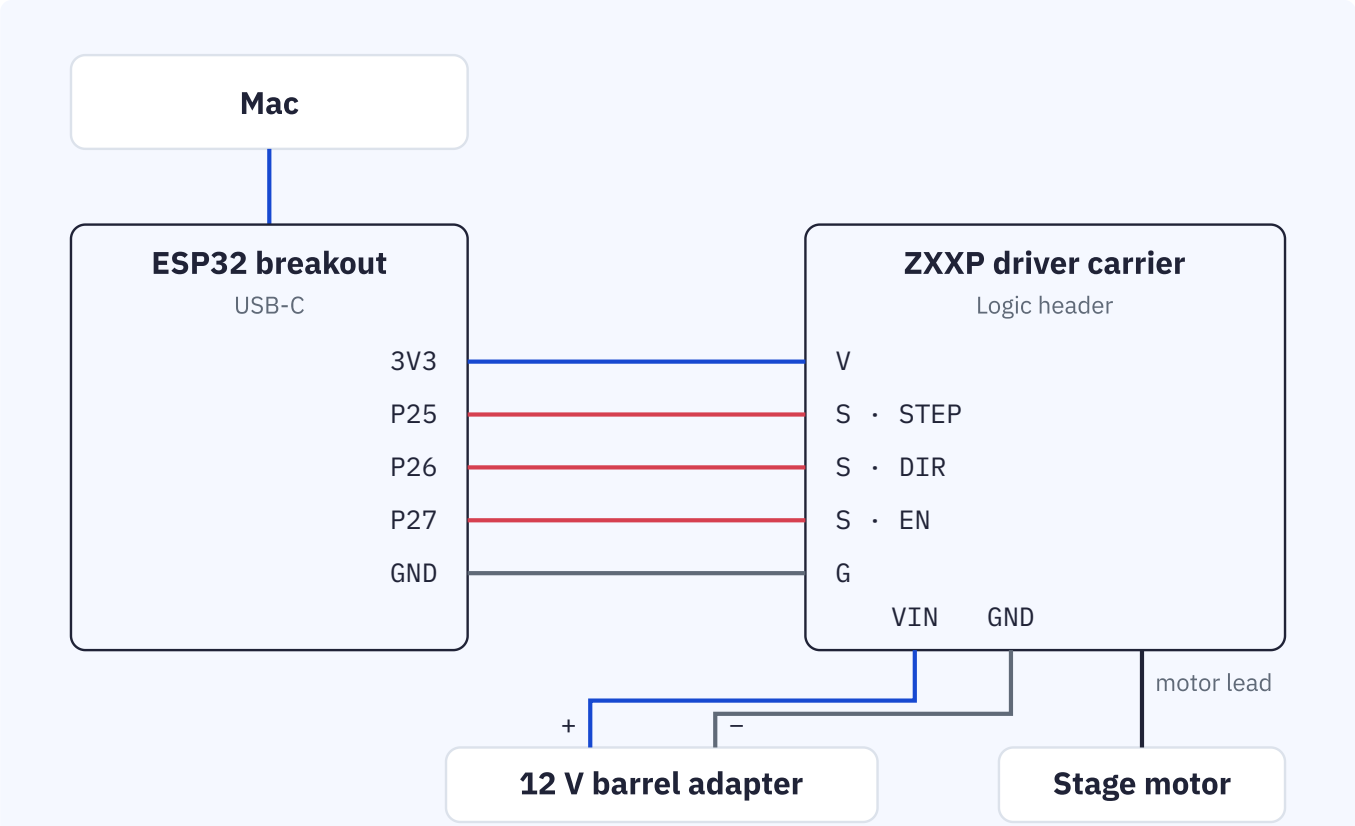
CHECK THE SOCKET

Match EN, STEP and DIR to their labeled signals. Match VCC_IO to logic supply, VS or VM to motor supply, and both GND pins to ground.

BIGTREETECH pinout and current adjustment

Wire the drive and supplies

USB-C powers the ESP32. The barrel adapter connects only to the driver carrier's motor power terminals.



On the carrier, S is the signal row. V and G carry logic power and ground. VIN and the adjacent GND take the motor supply.

1. With both supplies unplugged, wire the ESP32 terminals to the carrier as shown. Connect the barrel connector's marked plus to `VIN` and minus to the carrier's power `GND`.
2. Measure resistance at the loose motor plug to identify the coil pairs. Expect black with green and blue with red. One pair goes to `1A/1B` and the other to `2A/2B`.

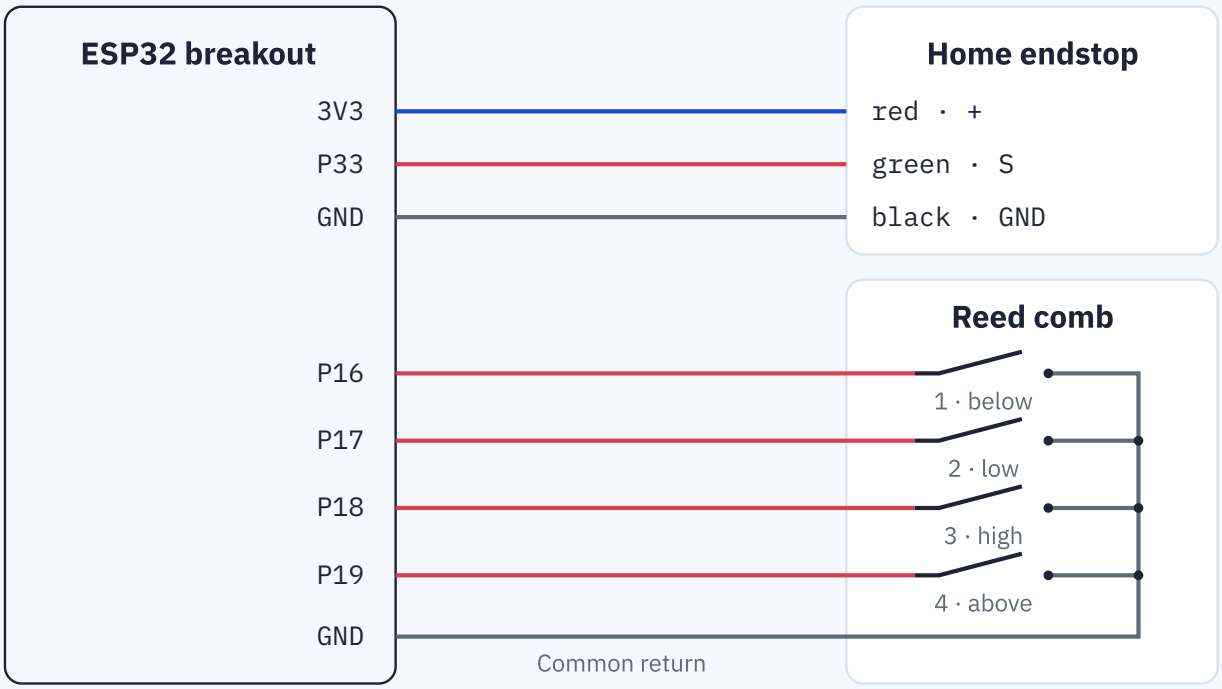
LEAVE POWER DISCONNECTED

Check each wire against the diagram. Leave the motor plug out through the current adjustment.

WIRE THE CONTROLLER

Wire the switch and reeds

The home switch takes power from the breakout. The reeds share a ground return and each has a dedicated input.



The ESP32 pulls each reed input up. A closed contact pulls its station input to ground.

1. Match the endstop lead colors against the board's markings. Run the red supply to 3V3, black return to GND and green signal to P33.
2. Connect the comb signals in station order from P16 through P19. Connect the common return to GND.
3. Route leads outside the cradle and away from the rod, carriage and coupling.

CHECK STATION ORDER

Trace each pocket back to its terminal from the tab end. The home lead and comb wires stay clear throughout the hand sweep.

CHECK THE WIRING

Meter the wiring before power

Disconnect USB and the barrel adapter. Leave the driver out of its socket and the motor unplugged while you check continuity.

PROBE AT	CHECK AT THE OTHER END
Barrel connector	Plus reaches carrier VIN. Minus reaches carrier power GND and the logic return.
Breakout 3V3	Reaches carrier V and endstop supply. On the empty carrier, V reaches the driver's VCC_IO socket.
Breakout GND	Reaches carrier G, endstop ground and reed return.
P25 / P26 / P27	Reach STEP / DIR / EN on the carrier and their respective driver sockets.
P33 and P16-P19	P33 reaches the switch signal. P16-P19 reach the comb leads in station order.
Unplugged motor	Measure resistance within each coil pair. The meter reads open between pairs.

CHECK FOR BRIDGES

Check VIN to GND, VIN to 3V3, and 3V3 to GND for shorts. Look for stray strands between terminals and tug each wire lightly.

CHECK THE CARRIER WITH MOTOR POWER ALONE

Unplug the carrier's logic header and leave the driver out. Connect the barrel adapter and verify positive supply voltage from VIN to power GND. Check V against G for voltage from the carrier. If V is powered, stop and identify its source before connecting the ESP32's 3V3. Unplug the adapter before reconnecting the logic header.

CHECK THE WIRING

Verify the inputs on USB

Leave the adapter and motor unplugged. Use the ESP32 console to check the reeds and home switch before the carriage can move.

Load the rig controller

Use `pio device list` to find the port that appears when you connect the ESP32 over USB. From the repository root, run these commands with that port in place of `<port>`.

```
pio run -e reed_rig -t upload --upload-port <port>
pio device monitor -e reed_rig --port <port> --echo
```

Check reeds and home switch

1. Lift the roller out of its slot and move the loaded sled by hand. Confirm that the console reports an `R` line at each closure and release, with the correct station number. Lower the roller back into its slot.
2. Send `status` with the lever free. Press and hold it, then send `status` again. The console should read `home free`, then `home pressed`.
3. If those states are reversed, read the pin level in `status` with the lever held. Send `homelevel low` for a low pin or `homelevel high` for a high pin. Repeat the lever check after any polarity change or reset, then leave the lever free.

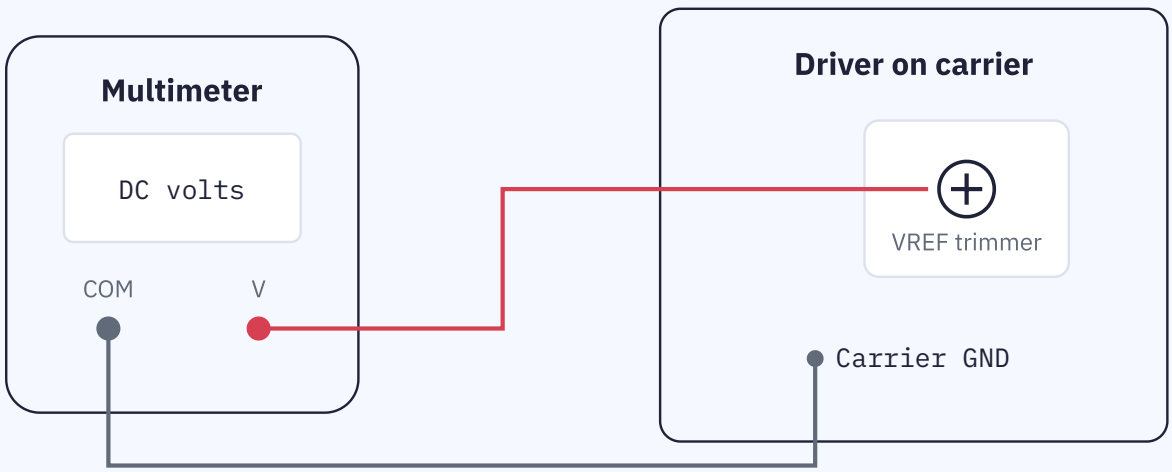
HOME SWITCH READY

With the lever released, `status` reports the home input free and the controller has registered a switch change since power-up.

CHECK THE WIRING

Set the drive current

Use the driver's reference voltage, VREF, to set the initial current limit. Leave the motor unplugged while you adjust it.



Clip COM to carrier ground before applying power. Touch the other probe to the driver's VREF point. Keep both probes clear of adjacent pins.

1. Disconnect both supplies. Seat the driver in the orientation shown on page 15. Verify the logic header is connected and clip the meter's COM lead to carrier ground.
2. Connect USB first, then the adapter. Measure VREF on the DC volts setting and adjust to about half a volt. Use an insulated tool and read the meter after each small adjustment.
3. Record VREF. Unplug the adapter and USB, then plug the motor lead into the carrier.

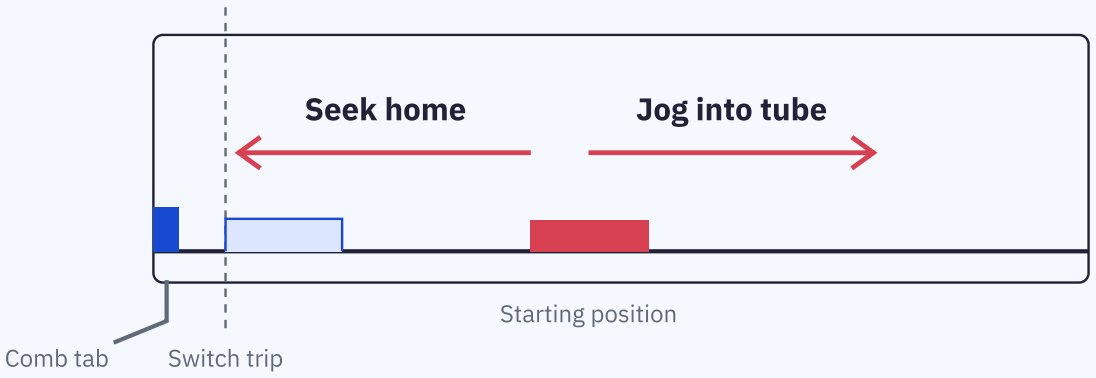
READY FOR A SHORT JOG

The measured VREF is set and recorded. Leave both supplies disconnected until the sled is positioned for the first jog.

RUN THE RIG

Verify direction and home

Start with the loaded sled clear of both ends and the roller in its slot. Keep the motor adapter within reach.



A positive jog moves into the tube. The switch trip sets home while there is still clearance at the comb tab.

1. Connect USB and open the console. Repeat the lever check on page 19, then connect the motor adapter. To stop a move, send `stop` in the console. If it keeps moving, disconnect motor power and reset the controller before repeating the lever check.
2. Send `jog 5`. If the sled moves toward the tab, stop. Use `status` to check the current direction, change it with `dirinvert on` or `dirinvert off`, then jog again.
3. Enter the measured card thickness in millimeters with `tab <mm>`. Send `home` and watch the sled approach the switch, back off, creep onto it, then back off to release it.
4. Home again to confirm that it stops at the same spot. Compare measured carriage travel with a short commanded jog.

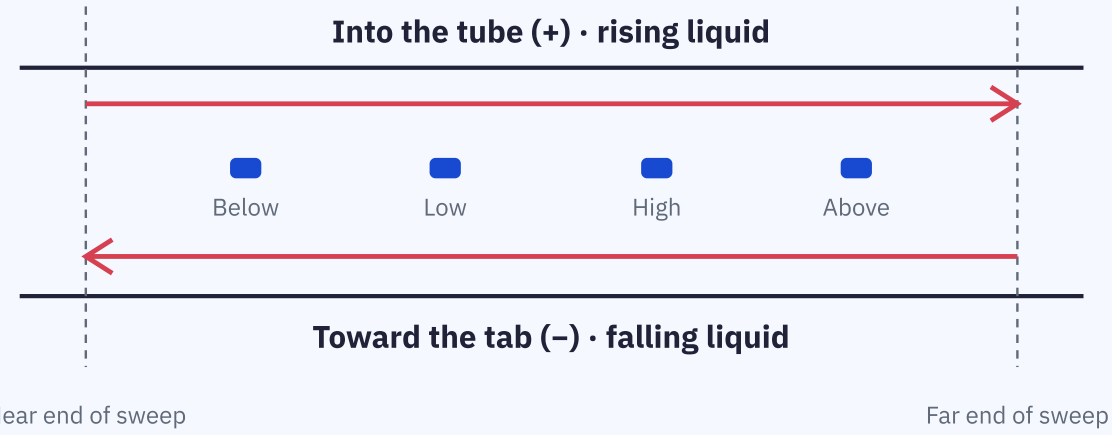
READY TO SWEEP

The switch trips before the sled reaches the tab and the carriage moves the commanded distance without binding. Home again after an interrupted move.

RUN THE RIG

Take a complete sweep

Choose endpoints that let the magnet pass every fitted reed in both directions.



Both endpoints must be within the controller's reach and far enough past the reed crossings for all fitted contacts to be open.

1. Clear loose steel. Try `sweep <from> <to>` in the console. Use millimeters from the tab to the sled's near face, within the range shown by `status`.
2. Close the console. From the repository root, run the command below with your port, sled, station assignments and measured values in millimeters. Use a new output directory for each run.
3. The log reader homes the sled and repeats the sweep. Press `Ctrl–C` to stop it. Keep the float and tube IDs and the measured wall gap with `raw.log` and `sweep.json`.

```
hsm-peripheral reed-sweep --port <port> --sled nominal \
  --reeds 2=A,3=B --play <roller-play> --glass <glass-length> \
  --from <from> --to <to> --output <new-run-directory>
```

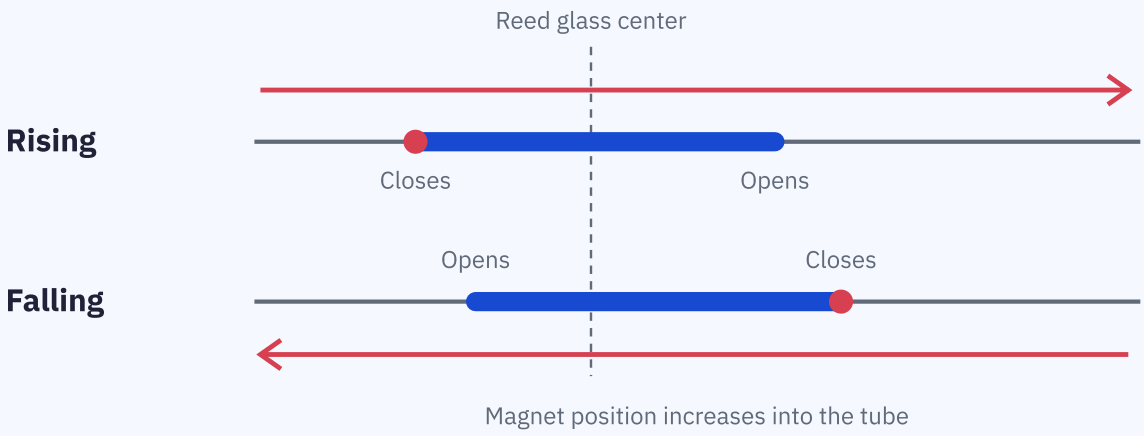
A COMPLETE RECORD

Check that each leg starts and ends in the log and every named reed has a complete open-close-open window in both directions. If an endpoint cuts off a window, adjust it within verified travel and rerun.

RUN THE RIG

Set reed heights from the sweep

Combine each reed's closure position with the float's measured waterline to calculate where the reed goes.



The blue spans show where the contact is closed. Follow the arrow for each leg to find the closure at the start of that leg's closed window.

1. Run NOMINAL, NEAR and FAR in separate directories. Check clearance and home again each time you change sleds.
2. For each reed, read the closure and release positions and their spread in `sweep.json`. Retain separate activation windows and keep `raw.log` for replay.
3. Use the rising closure for full (pump off) and the falling closure for empty (pump on). For each threshold, use the signed offsets below to calculate the reed center height. Record the result and the release position in the reed calibration record.

Reed center height = target liquid height + a − d

a = magnet center height minus liquid surface height, measured with the float in liquid.
d = magnet center height minus reed center height, from the directional closure in `sweep.json`.

CHECK IN LIQUID BEFORE FIXING THE REEDS

In the intended vessel, confirm both crossings repeat and occur before their respective travel stops. Apply the acceptance limits in the float calibration procedure before soldering fixed columns or printing the final bridge.

Rig procedure and log processing